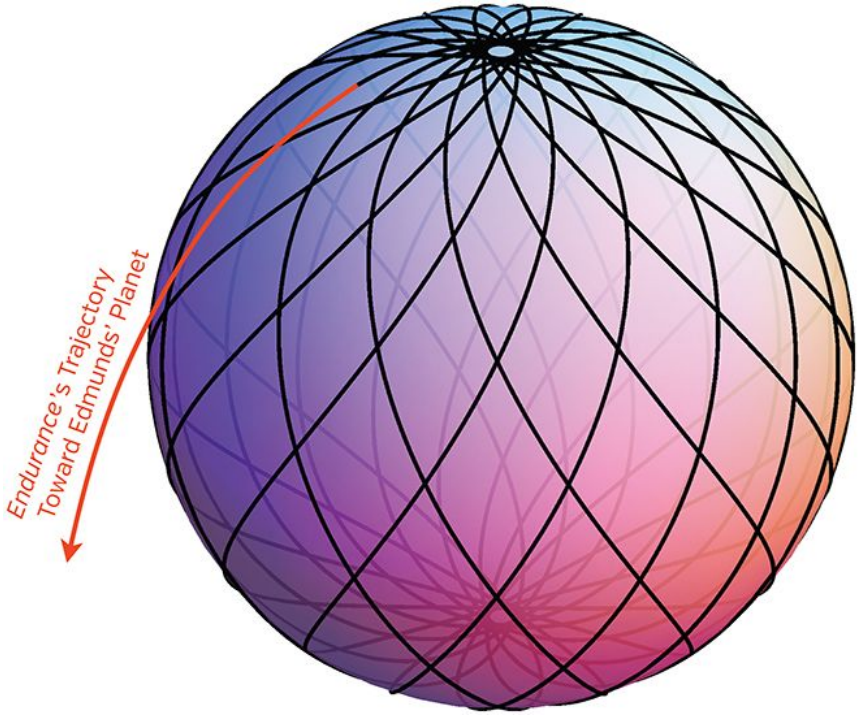


Fig. 27.10. A three-dimensional picture of the *Endurance*'s critical orbit and its launch toward Edmunds' planet. The critical orbit wraps around a sphere that surrounds Gargantua.

This convoluted critical orbit is a close analog of the trajectories of temporarily trapped light rays inside Gargantua's shell of fire (Figures 6.5 and 8.2). Like those light rays, the *Endurance* is temporarily trapped when on its critical orbit. Unlike the light rays, the *Endurance* has a control system and rockets, so its launch off the critical orbit is in Brand's and Case's hands. And because of the orbit's convoluted three-dimensional structure, the launch can be in any direction they wish.

But their launch leaves behind Cooper and TARS, plunging through Gargantua's horizon. Plunging toward Gargantua's singularities.

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49 This big difference is due to the *Endurance*'s having slightly less angular momentum than Mann's planet, after tidal forces have done their thing. In Figure 27.3 the *Endurance* climbs